

SAHAN VISHWA PERERA

SOFTWARE ENGINEER | TYPESCRIPT · WEBMCP · IOT | IT UNDERGRADUATE,
UNIVERSITY OF MORATUWA

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PROFESSIONAL SUMMARY

Software-engineering undergraduate at the **University of Moratuwa** (B.Sc. Hons IT & Management, 2025–2029) building **privacy-preserving WebMCP agent tooling (TypeScript/Vite/Node.js)** and **production IoT firmware (C++/ESP32)**. Shipped **Airlock** — in-browser spreadsheet analysis an AI agent queries without receiving the file (live demo) — and **WebMCP-Gauge** — lint + run harness (4,500 trials) measuring whether agents call exposed tools. Also sole engineer of an automated optical pH sensor (12-state FSM, custom PCB, WebSocket dashboard). Seeking a **software-engineering internship**.

EDUCATION

B.Sc. (Hons) in Information Technology & Management

2025 – Present

University of Moratuwa, Moratuwa

Expected Completion: 2029

G.C.E. Advanced Level

2023

St. Anthony's College, Kandy

ICT: A Accounting: A Economics: B

G.C.E. Ordinary Level

2020

St. Anthony's College, Kandy

IT: A English: A Religion: A + 6 more subjects

WORK EXPERIENCE

IT Technician (Internship)

Jan 2022 – Jun 2022

Kandy District Secretariat · Kandy, Sri Lanka

- Maintained **desktop hardware, OS imaging, and software** for a government IT environment — installs, troubleshooting, and backups across multiple departments
- Streamlined **Excel/Word data-entry and document workflows**, improving record accuracy and retrieval; diagnosed connectivity and config issues to cut downtime

TECHNICAL SKILLS

PROGRAMMING LANGUAGES

C C++ Java JavaScript TypeScript Node.js
SQL HTML5 CSS3

EMBEDDED SYSTEMS & IOT

ESP32 Microcontroller Programming
Firmware Development Finite State Machine (FSM)
PWM Control WebSocket LittleFS
PCB Design (EasyEDA) PlatformIO

WEB & AI-AGENT TOOLING

IT SUPPORT & SYSTEMS

WebMCP ToolsVitePapaParseREST APIs

Chrome CDPLLM Judge EvalsGitHub Actions

CLI Development

Hardware TroubleshootingOS Configuration

Network ConnectivityData Backup & Recovery

DEVELOPER TOOLS

GitGitHubVS CodeVersion Control

Microsoft Office Suite

SOFT SKILLS

Analytical Problem SolvingTechnical Communication

Self-directed Learning

CERTIFICATIONS

National Certificate — ICT Technician (NVQ Level 4)

2022

College of Technology, Kandy

PROJECTS

Airlock — Private In-Browser Spreadsheet Analyst (Devpost WebMCP Hackathon) — [GitHub](#) · [Live Demo](#)

TypeScript · Vite · WebMCP · PapaParse · HTML/CSS

- ▶ Built for the **Devpost WebMCP Hackathon challenge** — a **privacy-preserving data workbench** where CSV parsing and all aggregations run entirely in the browser tab — an AI agent queries 7 WebMCP tools (**describe_dataset, sum_by_category, filter_rows, monthly_trend, find_anomalies, top_expenses, clear_highlights**) and receives only aggregates while the network stays silent
- ▶ Designed a **default-on privacy gate** (file name + row details withheld, matches highlighted on the human's screen) with unit tests per tool; documented 5 empirically verified WebMCP API behaviors against Chrome 151

WebMCP-Gauge — Agent Tool-Invocation Measurement

Node.js 24 · JavaScript (ESM) · Chrome CDP · LLM Judge Evals · GitHub Actions

Harness — [GitHub](#)

- ▶ Shipped **lint (static manifest rules) + run (S sessions × R repeats, Wilson intervals, JSONL resume, honest Shields badges)** CLI plus a **Marketplace GitHub Action** with split exit codes separating "could not measure" from genuine regressions
- ▶ Published **16 runs / 4,500 trials**: clean manifest **99.3%** vs deliberately degraded twin **83.1%** (sum_by_category 60.0%, top_expenses 26.7%), proving the metric discriminates real description defects

Automated Optical pH Sensor for Hydroponic Systems — [GitHub](#)

C++ · ESP32 · PlatformIO · FSM · WebSocket · LittleFS · PCB Design (EasyEDA) · HTML/CSS/JavaScript

- ▶ Sole engineer of a **production-grade automated IoT colorimetric pH sensor** — embedded C++ firmware, custom PCB, 3D-printed optics enclosure, 3-pump fluidics, and real-time web dashboard from concept to completion
- ▶ Architected a **12-state non-blocking FSM** (fill → dose → mix → Dynamic Blanking → measure → vacuum drain) with dual-wavelength optics, TCS34725 HSV colorimetry, and NVS calibration for reliable hands-free operation